

Ismael Sierra

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Employment

University of Glasgow, (September 2026–)
Research Associate.

University of Toronto, (August 2023– July 2026)
Postdoctoral fellow.

Education

DPhil Mathematics, Cambridge University, (2019–2023)
Supervisor: Professor O. Randal-Williams.

MMath Mathematics, Cambridge University, (2015–2019)

Grants, Honours and Awards

Smith-Rayleigh and Knight-Rayleigh Price, Group 2, 2021.

Research Scholarship Trinity College Cambridge, 2019.

Publications

Ismael Sierra and Nathalie Wahl. Homological stability for symplectic groups via algebraic arc complexes. *Transactions of the American Mathematical Society*, 379(9), 2026.

Ismael Sierra. Homological stability of diffeomorphism groups of high dimensional manifolds via ek-algebras. *Advances in Mathematics*, 497:110994, 2026.

Ismael Sierra. Homological stability of spin mapping class groups and quadratic symplectic groups. *Forum of Mathematics, Sigma*, 14:e18, 2026.

Preprints

Alexander Kupers, Daniil Rudenko, and Ismael Sierra. On the surjectivity conjectures of Dupont and Monod. *arXiv preprint arXiv:2609.03963*, 2026.

Alexander Kupers, Daniil Rudenko, and Ismael Sierra. Mixed Tate motives over number fields. *arXiv preprint arXiv:2608.26392*, 2026.

Alexander Kupers, Daniil Rudenko, and Ismael Sierra. The Goncharov lie coalgebra of a field. *arXiv preprint arXiv:2606.23863*, 2026.

Talks, Conferences and Seminars

Mixed Tate motives over number (for an algebraic topologist)– 39th British Topology Meeting; Oxford, October 1-2, 2026.

Mixed Tate motives over number fields and Goncharov’s universality conjecture– Motivic Midlands conference; Nottingham, September 7-11, 2026.

The Goncharov program and homology of general linear groups– Algebraic topology seminar; Northeastern University, December 9, 2025.

The Goncharov program and homology of general linear groups– MIT’s algebraic topology seminar; MIT, December 8, 2025.

E_∞ homology of general linear groups and the Goncharov program– Perspectives on the cohomology of general linear groups; Münster, Germany, October 6-10, 2025.

The Rognes rank spectral sequence and Goncharov program via E_∞ -homology– Polylogarithms, homology of linear groups, and Steinberg modules; SwissMAP Research Station, Les Diablerets, Switzerland, June 9-13, 2025.

Homological stability of even (and odd) symplectic groups– Moduli and Friends Seminar; online, April 14, 2025.

Homological stability of symplectic groups– Geometry and Topology Seminar; Binghamton University, November 14, 2024.

Scissors congruences, algebraic K-theory and Steinberg modules; Pasadena, July 08-12, 2024.

Homological stability of even and odd symplectic groups– Geometric Topology Grad and Postdoc Seminar (GT GAPS); online, March 1, 2024.

Homological stability of even and odd symplectic groups– Geometry and Topology Seminar; University of Chicago, January 18, 2024.

Spaces of Manifolds: Algebraic and Geometric Approaches (23w5017); Banff, November 26- December 1, 2024.

Homological stability of even and odd symplectic groups– Geometry and Topology Seminar; McMaster University, September 28, 2023.

Homological stability of quadratic symplectic groups– Stability in Topology, Arithmetic, and Representation Theory; Purdue University, July 17-21, 2023.

Homological stability of moduli spaces via E_k -algebras– HHH follow on: Homotopy: fruit of the fertile furrow, Isaac Newton Institute; Cambridge, June 12-16, 2023.

Workshop on Homology of Configuration Spaces and related topics; Paris, May 16-23, 2023.

Homological stability of diffeomorphism groups of manifolds using E_k -algebras– Differential Geometry and Topology seminar, University of Cambridge, March 8, 2023.

Homological stability of diffeomorphism groups of manifolds using E_k -algebras– Geometry and Topology seminar, Warwick University, January 10, 2023.

Homological stability of diffeomorphism groups of manifolds via E_k -algebras– Geometry/Topology seminar University of Toronto (online), October 31, 2022.

Encuentro de Jóvenes Topólogos/ Encuentro de Topología; Zaragoza, October 18-22, 2022.

European Autumn School in Topology 2022; Utrecht, September 26-30, 2022.

Splitting complexes– Young Topologists Meeting; Copenhagen, July 18-22, 2022.

Homological stability of diffeomorphism groups and splitting complexes of manifolds– British Topology Meeting; Durham University, April 12-14, 2022.

Homological stability of diffeomorphism groups and splitting complexes of manifolds– Algebra/Topology seminar; University of Copenhagen, February 4, 2022.

Homological stability of diffeomorphism groups and splitting complexes of manifolds– Purdue Zoom topology seminar (online), September 22, 2021.

Oberwolfach Seminar: Cellular E_k -Algebras (hybrid meeting), May 23-29, 2021.

E_k -algebras and diffeomorphism groups– Young Topologists Meeting (online), July 5-16, 2021.

Heidelberg Laureate Forum 2017; Heidelberg, September 24-29, 2017.

Teaching Experience

Instructor (at University of Toronto)

Fall 2025 and Winter 2026 MAT137 Calculus with proofs.

Fall 2024 and Winter 2025 MAT137 Calculus with proofs.

Winter 2024 MAT 337 Introduction to real analysis.

Fall 2023 MAT 223 Linear Algebra I.

Supervisor (at Cambridge University)

IB Analysis and Topology, Michaelmas 2019.

II Algebraic Topology, Michaelmas 2019, revision Easter 2020, Michaelmas 2020, revision Easter 2021, Michaelmas 2021, revision Easter 2022, Michaelmas 2022.

IB Geometry, Lent 2020, revision Easter 2020, Lent 2021, revision Easter 2021, Lent 2022.

II Differential Geometry, Lent 2020, revision Easter 2020, Lent 2021, revision Easter 2021, Lent 2022, revision Easter 2022.